Technical approach

1 / Architecture and scope

Decision boundary

The prototype supports the Q4 2024 board cycle using public SEC filings dated on or before February 28, 2025, Dana's agreed preparation-date proxy. Reporting period and publication date remain separate. The supplied Apex file is disputed and withheld until a named approver and approval note are recorded; even after approval, rows after Q4 2024 cannot enter this historical comparison.

Data flow

Python CLI -> SEC submissions inventory, including historical pages -> eligible financial, portfolio, prospectus and current reports -> complete submissions and exhibits -> source cache -> structured facts and evidence candidates -> metric-specific resolution -> HTML, CSV, PDF and JSON. SEC sources use declared identification, a four-request-per-second ceiling, bounded retries and immutable SHA-256 content. Cached evidence retains its retrieval timestamp; offline runs cannot invent a response.

Extraction boundaries

BeautifulSoup/lxml parse inline XBRL, HTML tables and narrative. PDF text is extracted with pypdf. Selected NAV/book-value tags can qualify when units, instant date and undimensioned common-share context are explicit. Simple return tables qualify only with matching class/basis, annualization, reinvestment and date evidence. Narrative rules collect candidate rates and entire supporting passages. Complex tables, class dimensions and ambiguous narratives require recorded review. N-PORT is discovered and retained; dedicated XML aggregate mapping is not yet implemented.

Financial definitions

For Golub and KREF, the primary return is shareholder total return; interval funds use net NAV returns for the intended class. These bases remain visibly different. Contractual management fees are primary; disclosed effective fees remain separate, with waiver terms in evidence. KREF is a mortgage REIT: book value is not labeled fund NAV, and leverage/yield require their stated definitions. No undocumented gross-to-net, leverage or yield conversions are performed.

Implementation choice

A deterministic local pipeline avoids model credentials and makes rules testable. It favors explicit gaps over broad but unverified coverage. An HTML evidence queue makes source inspection practical; PDF and CSV travel with the quarterly package. This is an implementation default pending confirmation of the final client format, not a claim that the existing Excel template is required.

Technical approach

2 / Validation, confidence and reconciliation

Per-value evidence contract

Every candidate records ticker, metric, value if parsed, units, period, share class, basis, accession, filing date, original document URL, document hash, exact locator, source quote and extraction method. HTML uses source anchors, block/table ordinals or XBRL contexts instead of fabricated PDF page numbers. PDF passages use actual page numbers. Complete submissions and extracted exhibits are cached so results can be reproduced even if web presentation changes.

Confidence model

Confidence is evidence status, not a probability. Medium means structured unit/context/date/basis checks passed. Reviewed means a named human accepted the interpretation with a verbatim evidence excerpt and rationale. Neither signifies PM or compliance approval. Needs-review, conflict and unavailable states keep the displayed value blank. A separate comparability explanation describes why a correctly extracted figure may still differ conceptually from another fund's figure.

Conflicts and freshness

For a metric, select the latest eligible reporting period at or before target. Different accepted values or definitions for that same period are withheld rather than resolved by filing recency alone. Numeric unresolved candidates in the same or a newer filing also block publication. Explicit rejection requires reviewer identity and reason. Any retrieval failure marks fund coverage incomplete, preventing a stale value from appearing current. Period lag over 120 days is flagged as a configurable review threshold, not a business materiality policy.

Review controls

Decisions bind to source-derived candidate IDs. Acceptance requires a quoted substring, the numeric value present in that excerpt, compatible units, date, basis and class. Unsupported derived values cannot be manually introduced as extracted facts. Review is a local auditable file, not authenticated approval infrastructure. The output remains a draft until the PM and compliance independently approve it.

Validation performed and not performed

Tests use synthetic unit fixtures exclusively to exercise cutoff inclusion/exclusion, historical discovery, amendments/conflicts, units/scales, class ambiguity, evidence-bound decisions, internal-data withholding, and failure exports. Direct live SEC calls in development returned HTTP 403; no competitor facts or actual filing disagreements have been validated. It would be misleading to claim completed financial reconciliation. After access is restored, manually audit actual source passages, reconcile NAV with reported balances where definitions permit, inspect fee waivers and compare matching return periods before calibrating coverage.

Technical approach

3 / Limitations, risks and rollout

Live-access blocker

SEC rejected direct development requests with HTTP 403. The application records the source URL and error, outputs no invented values and exits nonzero for incomplete retrieval. The initial smoke run used the developer's GitHub project as contact; rerun with a real SEC_USER_AGENT email and permitted network. Do not treat this failure as fund non-disclosure or certify a populated output until real filings have been downloaded and audited.

Extraction and normalization risk

The generic rules deliberately cover fewer layouts than a production service. Narrative percentages can refer to expense ratios, old terms or thresholds; they remain candidates. Institutional class mappings need confirmation from actual filings. Scanned PDFs need OCR; merged tables need more fund-specific rules. There is no automatic market-price return calculation, general N-PORT normalization, or validated leverage derivation. Disclosed NAV/shareholder return divergence is not automatically reconciled; matching-period examples should be reviewed with Dana.

Source-selection risk

Broad historical report and recent supplement discovery reduces missed exhibits but can be expensive. Current metadata filtered by filing date is not a separately archived 2025 snapshot. Filing and amendment availability should be checked against the SEC index during calibration. Conservative fund-wide blocking after a failed document can reduce usable coverage; production should track which metrics depend on each missing source without silently weakening freshness claims.

Production safeguards

Add per-fund regression documents, extraction coverage measures, authenticated reviewer roles, immutable decision versions and a PM/compliance release gate. Keep a filing inventory diff on each run to detect new amendments. Store raw sources in versioned object storage, use a shared rate limiter and alert on SEC denials, source gaps and definition changes. Client inputs require access controls and approved provenance. Keep outputs private until authorized.

Delivery and next checkpoint

The repository supplies installation, CLI, local evidence review, portable exports, tests, decisions and the business brief. Complete the live pull first; use a representative sample of actual results to quantify correct extraction and review burden. Bring a real fee conflict and the KREF comparison to Dana in the next consultation. Confirm the output format and internal data before staging a PM review. Scheduling and hands-off operation should follow successful quarterly parallel runs, with acceptance criteria agreed by the client.